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LLM Inference and Serving Optimization | SLA Aware Routing | Cross Stack Performance | Agentic Systems

SUMMARY

AI systems researcher with 10 years of experience in LLM inference optimization, SLA aware routing, and large scale serving infrastructure. Lead author on PROTEUS, a Lagrangian RL based router that enforces joint latency and cost SLAs across heterogeneous model tiers, accepted as an **oral paper at EuroMLSys 2026** (co located with EuroSys, Edinburgh). Strong track record of identifying cross stack bottlenecks across runtime, framework, and serving layers and driving the optimizations end to end. This work has translated into a 4x inference cost reduction at Reliance Jio and a further 2.5x reduction at Quantiphi R&D across production multi tenant pipelines. 4 filed patents and an active member of the vLLM open source serving community.

HIGHLIGHTS

- **PROTEUS, an SLA aware LLM router** built on Lagrangian dual variable RL, accepted as an oral paper at EuroMLSys 2026 (Edinburgh, April 2026). Enforces joint latency and cost SLAs dynamically across heterogeneous model tiers and reduced production serving cost by 43%.
- **4x inference cost reduction at Reliance Jio** through systems level optimization of the LLaMA 3.1 8B serving pipeline (INT8 quantization, pruning, vLLM deployment) on a 4xA100 cluster, while holding p95 latency under 100 ms across 10K daily queries.
- **KV Hub, a cross model KV cache transfer layer** that eliminates redundant prefill computation when switching between model families in multi model serving. Patent filed. Achieved 35% prefill latency reduction in production pipelines.
- **4 filed patents** covering SLA aware LLM routing (PROTEUS), cross model KV cache architecture (KV Hub), and kinematics based sports injury prediction.
- **Active member of the vLLM community**, tracking and discussing developments in speculative decoding, paged KV cache, and serving efficiency, with that work feeding directly into the research and patents above.

AREAS OF EXPERTISE

LLM Routing and Serving Systems: SLA aware routing, Lagrangian dual variable RL, constrained MDPs, multi tenant SLO enforcement, capacity planning, queuing theoretic modeling, Pareto frontier analysis for cost and latency trade offs.

LLM Inference Infrastructure: vLLM, TensorRT-LLM, Triton Inference Server, TorchServe, ONNX, continuous batching, PagedAttention, prefill and decode disaggregation, speculative decoding (draft target, EAGLE), cross model KV cache reuse.

System Level Performance Optimization: Cross stack bottleneck analysis across runtime, framework, and hardware. Latency profiling at p50, p95, and p99, throughput characterization, end to end performance regression analysis.

Model Compression and Quantization: GPTQ, AWQ, INT8 and mixed precision, pruning, knowledge distillation, zero calibration quantization.

GPU and Heterogeneous Compute: CUDA, mixed precision (FP16, BF16, INT8), tensor and pipeline parallelism, FlashAttention, memory hierarchy tuning. Earlier work on Intel Arria FPGA hardware aware compression for vision workloads.

Frameworks and MLOps: PyTorch, TensorFlow, HuggingFace Transformers, Docker, Kubernetes, KubeFlow, MLFlow, FastAPI, Prometheus.

EXPERIENCE

AI Research Lead, LLM Efficiency and Optimization, Quantiphi R&D, Bengaluru

Feb 2025 – Present

- **Designed PROTEUS**, a Lagrangian RL routing system that enforces joint latency (p95 under 200 ms) and cost (under \$0.02 per 1K tokens) SLAs across LLaMA 3 8B and 70B and Mistral 7B and Mixtral 8x7B tiers. **Reduced serving cost by 43%** over static allocation. Accepted as an oral paper at EuroMLSys 2026.
- **Built KV Hub** for cross model KV cache reuse via terminal layer alignment, achieving a **35% prefill latency reduction** in multi model serving pipelines by eliminating redundant prefill computation on model switches.

- **Production agentic routing pipeline** with SLM based tool orchestration, delivering a **2.5x inference cost reduction** for customer support and finance assistants through intelligent model selection and caching.
- **Filed 2 patents** covering the KV Hub architecture and the PROTEUS dual variable SLA controller, with comprehensive prior art analysis.

Senior AI Scientist, Reliance Jio, Hyderabad

Dec 2021 – Jan 2025

- **Indic Language LLM Query Resolution.** Led a team of 6 AI engineers to build an in house LLM based query resolution system across Hindi, Marathi, and Telugu. Instruction tuned LLaMA 3.1 8B with SFT and LoRA on 150K conversational examples.
- **Inference stack optimization.** Drove the serving pipeline with INT8 quantization, structured pruning, and vLLM deployment on a 4xA100 cluster, achieving a **4x inference cost reduction** while holding p95 latency under 100 ms across roughly 10K daily queries.
- **AAAI Workshop publication.** Hybrid LLaMA 2 70B and CRF decoder for Hindi open domain IE, POS tagging, and sense disambiguation. Achieved a 2.5% F1 gain over baseline against StanfordCoreNLP, LLaMA 2 70B, and GPT-4o.

Data Scientist, Legato Health Technologies (Anthem Inc.), Hyderabad

Sep 2020 – Nov 2021

- **End to end medical transcription system** using NVIDIA NeMo with TitaNet embeddings and the Multi Scale Diarization Decoder (MSDD), achieving 95% speaker identification accuracy across 500+ hours of clinical conversations.
- **Transformer based NLP serving stack** with BERT for NER and sentiment, and T5 for summarization, deployed on TensorRT optimized serving. Reduced minutes of meeting generation time by 70%.

Deep Learning Software Engineer, Inspirit-IoT Technologies, Chennai

Jan 2019 – Aug 2020

- **DNNOptimizer (FPGA compression framework).** Collaborated with UIUC PhD researchers on a hardware aware DNN compression framework that surpassed TensorRT on ResNet-50, YOLOv3, and FaceNet running on Intel Arria FPGAs.
- **Cross stack kernel and pipeline tuning** for object detection and facial recognition workloads. Partnered with hardware architects on FPGA specific memory layout and compute optimizations.

Machine Learning Engineer, Virtusa Consulting, Hyderabad

Nov 2016 – Dec 2018

- **Production ML pipeline** processing 100K+ daily insurance claims with hybrid LSTM and GRU models at 87% accuracy, alongside real time fraud detection at 92% accuracy. Scalable microservices architecture reduced latency by 60% at 95% uptime.

EDUCATION

MS by Research (NLP), Data Science, Liverpool John Moores University

Oct 2020 – Jul 2022

Thesis: *Neuroticism Alignment Detection for Misogyny Aggression via Hierarchical Explanation Method (HEDGE + SHAP-based XAI).*

B.Tech, Robotics and Control Engineering, NIT Jalandhar

Jul 2012 – Jul 2016

PUBLICATIONS & PATENTS

- **PROTEUS, a Lagrangian RL based SLA aware LLM Router.** Oral presentation, EuroMLSys 2026, Edinburgh (April 2026).
- **Towards Open Domain Automatic Information Extraction in Hindi.** AAAI Workshop.
- Filed patents. **KV Hub** (cross model KV cache architecture), **PROTEUS** (dual variable SLA controller), **Triad Three Coordinate Kinematics** (India Patent No. 202311036108), **Kinematics Spatial Vision Grading for ACL Tear Prediction** (India Patent No. 202311036107).